

Jonathan M. Binder

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US citizen · UK settled status

EDUCATION

Royal Holloway, University of London

PhD, Biological Sciences

Egham, Surrey, UK

2022 - 2026 (expected)

- Fully-funded London Interdisciplinary Biosciences Consortium DTP (LIDo) student
- Researching microbiological effects on the longevity of agricultural grass weed *Alopecurus myosuroides* seeds in the soil seedbank
- Methods: microbiology, molecular biology, bioinformatics, germination modelling, machine learning, and biochemistry
- Mentorship: Mentored two final-year undergraduate students for their independent research projects
- Industry internship: 3-month internship to be completed in Q1 2026 at Syngenta, Jeallot's Hill, UK

Supervisors:

- Gerhard Leubner-Metzger, PhD - *Professor in Plant Biochemistry, Royal Holloway, University of London*
- Kazumi Nakabayashi, PhD - *formerly Research Assistant in Molecular Seed Biology at RHUL, currently Associate Professor at Obihiro University of Agriculture and Veterinary Medicine, Obihiro, Japan*
- Thomas Holloway, PhD - *Research Technical Lead, Syngenta, UK*
- Ben Oyserman, PhD - *Global Technical Manager Plant & Soil Health, Syngenta, Switzerland*

The Pennsylvania State University

MSc, Agronomy

State College, PA, USA

2016 - 2019

- Field-based research examined the effects of cereal rye management, dairy manure application method, and timing of fall manure application on nutrient uptake, corn yields, and total forage production
- Research on field-scale lysimeters explored the effects of cereal rye management on surface and subsurface nitrogen and phosphorus transport
- Coursework included statistics, chemistry, weed science, crop nutrient management, forage management

Supervisors:

- Heather Karsten, PhD - *Associate Professor of Crop Production/Ecology*
- Douglas Beegle, PhD - *Distinguished Professor of Agronomy*

Kyung Hee University, Global Campus

Korean Language Institute (한국어학당)

Suwon, Gyeonggi-do, South Korea

2014 - 2015

- Took classes 20 hours per week for one academic year
- Achieved Level 6 (out of 6) on the Test of Proficiency in Korean (TOPIK) (한국어능력시험)

Vassar College

BA, Environmental Studies

Poughkeepsie, NY, USA

2010 - 2014

- Concentrations in biology and geography
- Coursework included plant biology, genetics, and an experiential course on the American Grasslands
- Participated in a learning and living community on food and farming with an internship on a grassfed beef operation

PUBLICATIONS

In preparation

Binder, J.M., Nakabayashi, K., Holloway, T., Oyserman, B., and G. Leubner-Metzger. Soil carbon and microbial activity mediate germination of *Alopecurus myosuroides* seeds infected with pathogenic fungi. *In preparation to be submitted to Plant and Soil*, 2026.

Royo-Esnal, A., Nikolić, N., **Binder, J.M.**, et al. Climatic drivers of *Echinochloa crus-galli* seed base temperature and base water potential at continental scale. *In preparation to be submitted to Weed Research*, 2026.

Binder, J.M., Alonzo-Flores, I., Arévalo-Rodríguez, L.A., Leubner-Metzger, G., and R. Cifuentes. Spatial random forest analysis for fertilizer recommendation: Understanding yield limiting factors for Guatemalan cardamom. *In review at Geoderma Regional*, 2026.

Schaedler, C.E., Machado, Y.M.S., Hepp, S.S.H., **Binder, J.M.**, Felix, S.R., Langer, C.O., et al. Seed recovery, germination, and dispersal distances of herbicide-resistant *Amaranthus hybridus* and invasive *Eragrostis plana* by birds. *In review at Ecology & Evolution*, 2026.

In publication

Steinbrecher, T., Bhattacharya, S., **Binder, J.**, Kleemeier, K., Przesdzink, F., Groene, F., et al. Comparative pericarp biomechanics and germination physiology of *Raphanus raphanistrum* and *Raphanus pugioniformis* indehiscient fruits. *Annals of Botany*, mcaf015, 2025. <https://doi.org/10.1093/aob/mcaf015>.

Binder, J.M., Karsten, H.D., Beegle, D.B., and C.J. Dell. Winter annual management to increase nutrient recovery and forage production on dairies. *Agrosystems, Geosciences & Environment*, 4:e20157, 2021. <https://doi.org/10.1002/agg2.20157>.

Binder, J.M., Karsten, H.D., Beegle, D.B., and C.J. Dell. Manure injection and rye double cropping increased nutrient recovery and forage production. *Agronomy Journal*, 112, 2968–2977, 2020. <https://doi.org/10.1002/agg2.20181>.

COMMUNICATION

Oral presentations

EWRS Invasive Alien Plants and Biological Control Working Groups Workshop, Antalya, Türkiye, 2024
Northeast Cover Crop Council Annual Meeting, State College, PA, 2018
ASA, CSSA, SSSA Annual Meeting, Baltimore, MD, 2018
Pennsylvania Double-Cropping Winter Farmer Workshop, State College, PA, 2018

Poster presentations

15th Biennial International Seed Science Society Conference, Perth, Australia, 2025
5th Plant Microbiome Symposium, Amsterdam, Netherlands, 2024
International Seed Science Society Annual Meeting, Paris, France, 2023
British Crop Protection Council Weeds Review, Cambridge, UK, 2022
Northeastern Plant Pest and Soils Conference, Washington, DC, 2018
Northeast Cover Crop Council Annual Meeting, Ithaca, NY, 2017

ACADEMIC SOCIETY MEMBERSHIPS

Seed Microbiome Working Group
European Weed Research Society (EWRS)
International Seed Science Society (ISSS)
Tropical Agriculture Association International (TAAI)

FUNDING AWARDS

Global Challenges Research Fund

Mar 2025

- Funded by UK Research and Innovation (UKRI)
- £2000 awarded for networking trip to Guatemala to engage in trust-building, and brainstorming with collaboration researchers at Universidad de Valle de Guatemala
- Topics included soil health and fertilizer recommendations for cardamom production

LIDo Program PhD Studentship

Sept 2022- Sept 2026

- Awarded tuition waiver, stipend, laptop, and research funds totalling approx. £240,000
- One of a capped number of foreign students accepted to the program
- Funded by Biotechnology and Biological Sciences Research Council (BBSRC)/UKRI and Syngenta AG as industry partner

Penn State Graduate Research Studentship

2016-2019

- Awarded tuition waiver, stipend, and research funds valuing approx. \$100,000
- Part of a larger long-term, interdisciplinary dairy cropping experiment
- Funded by a USDA Sustainable Agriculture Research Education (SARE) grant and the Sahakian Family Charitable Foundation

Ann Cornelisen Fellowship for Language Study

2014-2015

- Awarded post-Graduate fellowship to study Korean in South Korea valuing approx. \$10,000
- Funding came from an endowment by a Vassar College alumna and Vassar seniors and alumni are eligible to apply

WORK EXPERIENCE

Royal Holloway, University of London

Jan 2024-Mar 2024

Module co-convenor

Egham, Surrey, UK

- Taught R programming language for pre-baccalaureate program biology students
- Constructed course materials and led computer lab sessions in statistical analysis and data visualization

NIAB East Malling

July 2021-June 2022

Research technician

East Malling, Kent, UK

- Collected and analyzed various data related to precision agriculture and novel sensors
- Verified and developed technologies in concert with startup companies
- Duties included tractor-mounted LiDAR scans of apple orchards, lab-based soil chemical assays, plant leaf and fruit imaging, and data analysis

NRI, University of Greenwich

Sept 2020-July 2021

Research technician

Chatham Maritime, Kent, UK

- Assisted senior researchers and PhD students with projects in entomology and agronomy departments at the Natural Resources Institute (NRI)
- Responsibilities included maintaining insect colonies, plants, greenhouses, and research facilities

Phospholutions

Mar 2019-May 2020

Research associate

State College, PA, USA

- Developed and researched efficacy of novel phosphorus fertilizers and phosphorus capture and recycle systems

- Responsibilities included literature review, in-house adsorption and desorption studies, creation of protocols, data analysis, and communication with industry and academic researchers

The Pennsylvania State University

Graduate research assistant

Sept 2016-Mar 2019
State College, PA, USA

- Conducted original research for master's thesis on cereal rye and manure management for dairies in the Northeastern U.S.
- Collected and analyzed soil and crop data, scouted for pests, and presented research findings in academic and industry settings as part of a larger long-term, sustainable dairy cropping system project

Hudson Valley Research Laboratory

Research technician

Sept 2015-Aug 2016
Highland, NY, USA

- Assisted with research at an entomology lab primarily focused on apple pests
- Tasks included fruit damage assessments, microscopy, scouting, and insect rearing as well as upkeep of the research orchard and facility
- Pruned, mowed, managed weeds, and ran tractors and a skid steer
- Completed light construction, plumbing, and tractor maintenance

Summer Hope Academic Research Program

Undergraduate researcher

June 2012-July 2012
Holland, MI, USA

- Participated in an undergraduate research program at Hope College
- Worked in the lab of Dr. Jianhua Li on a maple tree phylogeny project
- Techniques included chloroplast isolation, DNA extraction, and PCR

Vassar College Biology Department Greenhouse

Greenhouse assistant

Sept 2011-May 2014
Poughkeepsie, NY, USA

- Federal Work-Study recipient (need-based, part-time employment program)
- Maintained glasshouse and plant collection
- Duties included watering, fertilising, repotting, sweeping, and weeding

Vassar College Grounds Services

Student worker

June 2011-August 2011
Poughkeepsie, NY, USA

- Planted and maintained flower beds and gardens on campus
- Duties included soil preparation, mulching, watering, weeding, and fertilising

Poughkeepsie Farm Project

Seed saving garden assistant

Sept 2010-May 2011
Poughkeepsie, NY, USA

- Federal Work-Study recipient
- Worked in the seed saving garden at a community-supported agriculture (CSA) operation
- Duties included collecting and processing vegetables for seed saving and garden work

SHORT COURSES TAKEN

Nanopore metagenomics

Earlham Institute

July 9-10, 2024
Norwich, UK

- Short course on isolating, sequencing, and analysing metagenomic DNA
- Focused on the use of Oxford Nanopore Technologies sequencing devices
- Lectures by Drs. Richard Leggett, Darren Heavens, and Ned Peel

Seed germination modelling*European Weed Research Society (EWRS)*

June 13-15, 2023

Perugia, Italy

- Short course on using **dose response curve time-to-event** R package (**drctc**)
- Course presented by package author Prof. Andrea Onifri, University of Perugia

Weed genomics*EWRS*

Feb 27 - March 2, 2023

Antalya, Türkiye

- Included lectures by Drs. Roland Beffa (Independent consultant, Germany), Eric Patterson (Michigan State University), Dana MacGregor (Rothamsted Research), Maor Matzrafi (Volcani Institute, Israel)
- Hands-on bioinformatics training session

VOLUNTEERING**Durae Sandeul Farm** (두레산들 농장)*Farm volunteer*

June 2013-Aug 2013

Hwaseong-si, Gyeonggi-do, South Korea

- Homestayed on an organic farm through WWOOF Korea
- Activities included weeding, watering, planting, and harvesting in different vegetable cropping systems according to Korean natural farming (KNF - 한국식 자연농법) practices
- Participated in cultural activities including workshops in traditional music and crafts

SKILLS**Computer** - R, Quarto, Microsoft Office, Git, Bash, $\LaTeX$ **Soft** - Problem-solving, flexibility, critical thinking, resourceful**Dexterous** - Woodworking, plumbing, electrical**Qualifications** - Safe Handling & Application of Pesticides (PA1), Hand Held Applicators on Land & Near Water (PA6AW)**Languages**

- English - First language
- Korean - Professional working proficiency
- Spanish - Beginner
- French - Beginner